# Frontiers of Physical Chemistry (FPC 2026): Experiments, Computations and Theory

28 – 31 October 2026 | S. N. Bose National Centre for Basic Sciences, Kolkata

SBO / MCO: Named Oration Lecture, 30 minutes (25+5)
EL: Special Evening Lecture, 60 minutes (45+15) · CL: Commemorative Lecture, 60 minutes; both online
IL: Invited Lecture, 20 minutes (17+3)
SL: Invited Lecture (short format), 15 minutes (12+3)
IP: Invited Faculty Poster · P: Contributed Poster · SP: Sponsor Talk, 10 minutes

## DAY 1 — Wednesday, 28 October 2026

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| *08:00 – 09:00* | | *Registration and distribution of conference kits* |
| *09:00 – 09:15* | | **Inaugural Session** |
| **SESSION 1 (Chair: Swati De (University of Kalyani))** | | |
| 09:15 – 09:35 | **IL-01** | E. Arunan (IISc Bengaluru)<br>*Is the distance between two atoms enough to conclude about bonding?* |
| 09:35 – 09:55 | **IL-02** | Sarika Maitra Bhattacharyya (CSIR-NCL Pune)<br>*Beyond Crystallization Frustration: Structure and Dynamics in Model Glass Former* |
| 09:55 – 10:15 | **IL-03** | Pradipta Bandyopadhyay (JNU New Delhi)<br>*Generalizing Physics-based Machine Learning for prediction of hydration free energies* |
| 10:15 – 10:30 | **SL-01** | Srestha Basu (SINP Kolkata)<br>*Engineering Chemical Reactivity and Optical Response through Defect Design in Nanocluster Assemblies* |
| *10:30 – 11:00* | | *Tea Break* |
| **SESSION 2 (Chair: Sumana Dutta (IIT Guwahati))** | | |
| 11:00 – 11:20 | **IL-04** | Moloy Sarkar (NISER Bhubaneswar)<br>*From Surface Chemistry to Excited-State Reactivity: Unraveling Catalysis in Mono- and Bimetallic Nanoclusters* |
| 11:20 – 11:40 | **IL-05** | Durba Roy (BITS Pilani, Hyderabad)<br>*What Nonequilibrium Work Theorems Tell Us About Carbohydrate Mechanics* |
| 11:40 – 12:00 | **IL-06** | Partha Hazra (IISER Pune)<br>*TADF-Enabled Triplet Exciton Harvesting in Novel Organic Multi-functional Luminogens* |
| 12:00 – 12:20 | **IL-07** | Ankan Paul (IACS Kolkata)<br>*Shedding Light on the Mechanism of Highly Selective Toluene Photo-oxidation under Nano-Confinement* |
| 12:20 – 12:35 | **SL-02** | Saumyak Mukherjee (IISER Berhampur)<br>*A Bayesian Lens on Molecular Rigidity: Learning Atomic Weights from Ensemble Data* |
| 12:35 – 12:45 | **SP-1** | Sponsor Talk — Somananda Sanyal (Wiley-VCH) |
| *12:45 – 14:00* | | *Lunch* |
| **SESSION 3 (Chair: Pratim Chattaraj (SNBNCBS Kolkata))** | | |
| 14:00 – 14:20 | **IL-08** | Snehasis Daschakraborty (IIT Patna)<br>*$CO_2$ Adsorption in Humid COFs: A Transition Matrix Monte Carlo Study* |
| 14:20 – 14:40 | **IL-09** | Ayan Datta (IACS Kolkata)<br>*Buckling and Defects in Two-Dimensional Atomically Thin Monolayers* |
| 14:40 – 15:00 | **IL-10** | Debashis Barik (University of Hyderabad)<br>*Power-law scaling of depth of the potential wells in bistable systems* |
| 15:00 – 15:15 | **SL-03** | Supratik Sen Mojumdar (IIT Palakkad)<br>*Small-Molecule-Directed Self-Assembly of Amino Acids for Emergent Fluorescence* |
| *15:15 – 15:45* | | *Tea Break* |
| **SESSION 4 (Chair: Rajarshi Chakrabarti (IIT Bombay))** | | |
| 15:45 – 16:05 | **IL-11** | Srabani Taraphder (IIT Kharagpur)<br>*Free Energy and Kinetics of Unbinding of Enzyme Inhibitors* |

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| 16:05 – 16:25 | **IL-12** | Debasis Koley (IISER Kolkata)<br>*In Silico mechanistic rationalization of Copper-complex catalyzed oxidative reactions* |
| 16:25 – 16:40 | **SL-04** | T. Rajagopala Rao (IIT Patna)<br>*Quantum Nuclear Dynamics: Reactive Scattering and Vibronic Coupling* |
| **SESSION 5 — POSTER SESSION A (Chair: Surajit Maity (IIT Hyderabad))** | | |
| *16:45 – 18:15* | | *Invited faculty posters: IP-01 to IP-07*<br>*Contributed posters: P-001 to P-033, P-035, P-036* |
| *18:15 – 18:30* | | *Break* |
| **SESSION 6 — SPECIAL EVENING LECTURE (online) (Chair: Anindya Datta (INST Mohali))** | | |
| 18:30 – 19:30 | **EL-1** | Biman Bagchi (IISc Bengaluru) — online<br>*From Molecular Chirality to Living Helices: Symmetry Breaking, Cooperativity, and Handedness Selection* |
| *19:30* | | *Dinner (SNBNCBS)* |

## DAY 2 — Thursday, 29 October 2026

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| **SESSION 7 (Chair: Anunay Samanta (University of Hyderabad))** | | |
| 09:00 – 09:30 | **SBO** | **Sadhan Basu Oration Lecture**<br>Amalendu Chandra (IIT Kanpur)<br>*Nonlinear Vibrational Spectroscopic Calculations of the Dynamics of Aqueous Solutions at Charged Interfaces* |
| 09:30 – 09:50 | **IL-13** | Varadharajan Srinivasan (IISER Bhopal)<br>*An analytical hydrodynamic model for coupled edge plasmons in TMD nanoflakes* |
| 09:50 – 10:10 | **IL-14** | Parbati Biswas (University of Delhi)<br>*Rheology of Block Copolymeric Ring Polymers in Dilute Solutions* |
| 10:10 – 10:30 | **IL-15** | Vivek Tiwari (IISc Bengaluru)<br>*What is the electronic character evolution in singlet fission dimers?* |
| *10:30 – 11:00* | | *Tea Break* |
| **SESSION 8 (Chair: Rama Kant (University of Delhi))** | | |
| 11:00 – 11:20 | **IL-16** | Aloke Das (IISER Pune)<br>*β-Turn Peptides: Bridging Molecular Structure and Functional Materials* |
| 11:20 – 11:40 | **IL-17** | Arnab Mukherjee (IISER Pune)<br>*Memory Effects and Ruggedness as Microscopic Sources of Internal Friction in Chemical and Biophysical Dynamics* |
| 11:40 – 12:00 | **IL-18** | Prasun Kumar Mandal (IISER Kolkata)<br>*Does Excitation Energy Transfer Happen in Doped Perovskite Nanocrystal?* |
| 12:00 – 12:15 | **SL-05** | Mainak Sadhukhan (IIT Kanpur)<br>*A systematic Green's function formalism for kinetic energy density* |
| 12:15 – 12:30 | **SL-06** | Anirban Mondal (IIT Gandhinagar)<br>*From Descriptors to Discovery: A Physics-Informed Inverse Design Framework for Self-Trapped Excitons in Unexplored Halide Perovskites* |
| 12:30 – 12:45 | **SL-07** | Partha Pratim Roy (IIT Gandhinagar)<br>*Ultrafast Mapping of Nonradiative and Radiative Pathways in Dual-Conformer TADF Emitters* |
| *12:45 – 14:00* | | *Lunch* |
| **SESSION 9 (Chair: Sanjay Kumar (IIT Madras))** | | |
| 14:00 – 14:15 | **SL-08** | Arijit Kumar De (IISER Mohali)<br>*Unveiling Time-Dependent Vibronic Coupling by Third- and Fifth-order Raman Spectroscopy* |
| 14:15 – 14:30 | **SL-09** | Madhurima Jana (NIT Rourkela)<br>*Hydrophobic Amino Acid Solutions at Work: Do They Reshape Protein Solvation and Thermal Stability?* |

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| 14:30 – 14:45 | **SL-10** | Barun Kumar Maity (SINP Kolkata)<br>*Specific DNA Recognition Induces Folding Transition of the Oncogenic Fusion Protein PAX3FOXO1* |
| 14:45 – 15:00 | **SL-11** | Debasish Mondal (IIT Tirupati)<br>*Reset Controlled Non-equilibrium Transition in an Inhomogeneous Brownian Information Engine* |
| 15:00 – 15:15 | **SL-12** | Somrita Ray (IISER Berhampur)<br>*Stochastic Resetting Prevails Over Sharp Restart for Broad Target Distributions* |
| **SESSION 10 (Chair: Sabyashachi Mishra (IIT Kharagpur))** | | |
| 15:15 – 15:35 | **IL-19** | Susmita Roy (IISER Kolkata)<br>*Ions as Molecular Architects Shaping RNA Folding* |
| 15:35 – 15:55 | **IL-20** | Himansu Sekhar Biswal (NISER Bhubaneswar)<br>*From Conformer Resolution to Bond Scission: An Integrated Approach Using REMPI, RIDIR, and VCD/ECD Spectroscopy* |
| 15:55 – 16:15 | **IL-21** | Sanjoy Bandyopadhyay (IIT Kharagpur)<br>*Modified Interfacial Environment around Biomolecules in Aqueous Ionic Liquid Mixtures* |
| **SESSION 11 — POSTER SESSION B (Chair: Niladri Patra (IIT (ISM) Dhanbad))** | | |
| *16:15 – 17:45* | | *Invited faculty posters: IP-08 to IP-17, IP-27*<br>*Contributed posters: P-037 to P-069, P-105, P-106* |
| *17:45 – 18:00* | | *Break* |
| **SESSION 12 — SPECIAL EVENING LECTURE (online) (Chair: Amalendu Chandra (IIT Kanpur))** | | |
| 18:00 – 19:00 | **EL-2** | Sudipta Maiti (BITS Pilani, Hyderabad) — online<br>*Deep tech in photonics needs talent, ideas, resources, AND deep experience: let's put them together* |
| **19:30** | | **Conference Dinner — Hotel The Sojourn** |

## DAY 3 — Friday, 30 October 2026

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| **SESSION 13 (Chair: Sourav Pal (Ashoka University))** | | |
| 09:00 – 09:30 | **MCO** | **Mihir Chowdhury Oration Lecture**<br>Tapas Chakraborty (MAKAUT, West Bengal)<br>*Mass Spectrometry in Physical Chemistry Research* |
| 09:30 – 09:50 | **IL-22** | Rajesh Murarka (IISER Bhopal)<br>*Understanding the coil-globule transition in semi-flexible polymers from an energy landscape perspective* |
| 09:50 – 10:10 | **IL-23** | Ranganathan Subramanian (IIT Patna)<br>*Atmospheric Oxidation Kinetics of Emerging Halogenated Oxygenated Volatile Organic Compounds (XOVOCs)* |
| 10:10 – 10:30 | **IL-24** | Anirban Misra (University of North Bengal)<br>*Theoretical Design and Electronic-Structure Engineering of Donor–Acceptor π-Conjugated Molecular Systems for Tunable Nonlinear Optical Properties* |
| *10:30 – 11:00* | | *Tea Break* |
| **SESSION 14 (Chair: V. Subramanian (SRMIST Chennai))** | | |
| 11:00 – 11:20 | **IL-25** | Mousumi Das (IISER Kolkata)<br>*Role of excited states in optoelectronics: A model exact and DMRG study* |
| 11:20 – 11:40 | **IL-26** | Hemant Kumar Kashyap (IIT Delhi)<br>*Insights Into Antibacterial Action of Transition Metal Dichalcogenides* |
| 11:40 – 12:00 | **IL-27** | Tushar Debnath (Shiv Nadar University)<br>*Exciton-Plasmon Coupling in Plasmonic Perovskite Nanocrystals* |
| 12:00 – 12:20 | **IL-28** | Sanjib Senapati (IIT Madras)<br>*Reinventing the Drug Formulation Technology for Improved Onset of Action and Lower Dose Requirements* |

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| 12:20 – 12:35 | **SL-13** | Arnab Halder (Presidency University, Kolkata)<br>*Role of Photoinduced Processes to Modulate the Optical Properties of Carbon Nanoparticles* |
| 12:35 – 12:45 | **SP-2** | Sponsor Talk — Laser Science Services (India) |
| *12:45 – 14:00* | | *Lunch* |
| **SESSION 15 (Chair: Sanjib Bagchi (RKM Vivekananda Centenary College, Rahara))** | | |
| 14:00 – 14:20 | **IL-29** | Biman Jana (IACS Kolkata)<br>*Learning Transferable Collective Variables For Ligand Binding Transition Through Explainable Machine Learning* |
| 14:20 – 14:40 | **IL-30** | Manik Pradhan (SNBNCBS Kolkata)<br>*Cavity Ring-down Spectroscopy in Gas and Condensed Phases* |
| 14:40 – 14:55 | **SL-14** | Tatini Rakshit (Shiv Nadar University)<br>*Probing cancer biomarkers using a biophysical approach* |
| 14:55 – 15:10 | **SL-15** | Tapta Kanchan Roy (University of Delhi)<br>*An Efficient Variational Approach for Thermal Density Matrices with Quantum Anharmonic Effects* |
| 15:10 – 15:25 | **SL-16** | Atanu Singha Roy (NIT Meghalaya)<br>*Protein Corona Modulated Doxorubicin Loading and Release from Catechin-Functionalized Ag–Au Alloy Nanoparticles* |
| *15:25 – 15:45* | | *Tea Break* |
| **SESSION 16 (Chair: Avisek Das (IACS Kolkata))** | | |
| 15:45 – 16:00 | **SL-17** | Samiran Sen (Institut Pasteur, Paris)<br>*Learning what to trust: A Bayesian framework for protein model refinement* |
| 16:00 – 16:15 | **SL-18** | Brindaban Modak (BARC Mumbai)<br>*Density Functional Theory Study of Single-Atom Catalysts on Porphyrin-like Porous Fullerene for Energy Applications* |
| 16:15 – 16:30 | **SL-19** | Anupam Bera (University of Hyderabad)<br>*Molecular Structure and Ultrafast Charge-Transfer Dynamics at Functional Material Interfaces: Insights from VSFG and Transient Absorption Spectroscopy* |
| 16:30 – 16:45 | **SL-20** | Suman Das (GITAM University, Visakhapatnam)<br>*Proteins in Deep Eutectic Solvents: Molecular Insights from Simulations* |
| **SESSION 17 — POSTER SESSION C (Chair: Debabrata Seth (IIT Patna))** | | |
| *16:45 – 18:15* | | *Invited faculty posters: IP-18 to IP-22, IP-24 to IP-26*<br>*Contributed posters: P-070, P-072 to P-089, P-091 to P-104* |
| *18:15 – 18:30* | | *Break* |
| **SESSION 18 — SPECIAL EVENING LECTURE (online) (Chair: Ranjit Biswas (SNBNCBS Kolkata))** | | |
| 18:30 – 19:30 | **EL-3** | Mark Maroncelli (Penn State University, USA) — online<br>*Adventures of a Lonely Water Molecule as Told by IR, FIR, NMR Spectroscopy and MD Simulation* |
| *19:30* | | *Dinner (SNBNCBS)* |

## DAY 4 — Saturday, 31 October 2026

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| **SESSION 19 (Chair: Ranjan Das (West Bengal State University))** | | |
| 09:00 – 09:20 | **IL-31** | Musharaf Ali (BARC Mumbai)<br>*Thermophysical and dynamical properties of liquid sodium: Insights from MD simulations and DFT-BTE theory* |
| 09:20 – 09:40 | **IL-32** | Sayan Bagchi (CSIR-NCL Pune)<br>*Water Dynamics in Biomolecular Condensates: Beyond Slow Biomolecular Diffusion* |
| 09:40 – 10:00 | **IL-33** | Milan Hazra (IIT Jodhpur)<br>*Decoding Molecular Grammar of Heterotypic Biomolecular Condensates: Interplay of Sequence Architecture and RNA-like polyelectrolyte Mediated Competitive Interactions* |

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| 10:00 – 10:15 | **SL-21** | Biswajit Guchhait (Shiv Nadar University)<br>*Domain-Specific Dynamics and Solvent-Controlled Excited-State Proton Transfer in Deep Eutectic Solvents* |
| 10:15 – 10:30 | **SL-22** | Rakesh S. Singh (IISER Tirupati)<br>*From Interactions to Architecture: Rational Design of Biomolecular Condensate Morphologies* |
| *10:30 – 11:00* | | *Tea Break* |
| **SESSION 20 (Chair: Chaitali Mukhopadhyay (University of Calcutta))** | | |
| 11:00 – 11:20 | **IL-34** | Pradipta Purkayastha (IISER Kolkata)<br>*Epitaxial Assembly of 1D/2D Organic Lateral Heterojunctions for Integrated Photonics* |
| 11:20 – 11:40 | **IL-35** | Rahul Sharma (St. Xavier's College (Autonomous), Kolkata)<br>*Transient collective motion and density voids in a model drying chemotactic bacterial film* |
| 11:40 – 11:55 | **SL-23** | Satyapriya Bhandari (Kandi Raj College)<br>*Engineered Nanoarchitectures for Enhanced Energy Transfer and Multifunctional Applications* |
| 11:55 – 12:10 | **SL-24** | Piyush Mishra (IIT Kharagpur)<br>*Understanding Chemical Mechanisms Through Molecular Spectroscopy* |
| 12:10 – 12:20 | **SP-3** | Sponsor Talk |
| **SESSION 21 — Commemorative Session Remembering the Life and Works of Prof. B. M. Deb (Chair: Pratim Chattaraj (SNBNCBS Kolkata))** | | |
| 12:20 – 13:20 | **CL** | Swapan K. Ghosh (UM-DAE Centre for Excellence in Basic Sciences, Mumbai) — online |
| *13:20 – 13:50* | | **Valedictory Session** |
| *13:50 – 14:50* | | *Lunch and departure* |

## POSTER PRESENTATIONS

### POSTER SESSION A — Day 1, Wednesday 28 October 2026, 16:45 – 18:15 (Session 5)

| No. | Presenter | Affiliation | Title |
|---|---|---|---|
| **IP-01** | Bidyut Sarkar | Shiv Nadar University | Investigation of conformational dynamics and ligand binding mechanism of gene regulatory SAM-II / SAM-V tandem riboswitch |
| **IP-02** | Niladri Patra | IIT (ISM) Dhanbad | Mapping Conformational and Kinetic Landscapes of 14-3-3σ/α-Synuclein Assembly via Variational Autoencoders and Markov State Models |
| **IP-03** | Saumyak Mukherjee | IISER Berhampur | A Bayesian Lens on Molecular Rigidity: Learning Atomic Weights from Ensemble Data |
| **IP-04** | Debdas Dhabal | IIT Guwahati | CMA-ES-based Force Field Parameter Optimization of Diiodomethane for Surface Free Energy Calculations |
| **IP-05** | Pradip K. Ghorai | IISER Kolkata | Breaking Azeotropes: The Role of Entrainers in Efficient Separation |
| **IP-06** | Papri Chakraborty | IIT Kharagpur | Activation of atomically precise monolayer-protected Pt nanoclusters: Formation of active CO oxidation catalysts |
| **IP-07** | Utpal Sarkar | Assam University, Silchar | Machine Learning Approach to Screen Potential Donor and Acceptor Fragments: Insights from Chemical Reactivity |
| **P-001** | Hirak Jyoti Baishya | IIT Guwahati | Targeting Plasmodium Falciparum Purine Nucleoside Phosphorylase (PfPNP) with Unnatural Nucleosides: A Computational Study |
| **P-002** | Surendra Nath Barman | NIT Meghalaya | BSA Conjugated Carbon Nanotubes for Release of Potent Antioxidant Curcumin |
| **P-003** | Keniya Basu | University of Calcutta | On the origin of the structural isomerism dependent intersystem crossing : role of spin–orbit and nuclear motion assisted spin–orbit coupling |
| **P-004** | Saswata Basu | IIT Kanpur | Quantum phase transitions in time-dependent quantum mechanical scattering |
| **P-005** | Suman Bhowmik | IIT Kanpur | Decoding Intersystem Crossing: Molecular Design, Vibrational Control and Environmental Modulation |
| **P-006** | Partha Pratim Borah | IIT Guwahati | Rethinking Activity Descriptors for Single-Atom Iron Water Oxidation Catalysts |
| **P-007** | Roohi Choudhury | NIT Meghalaya | Deciphering bio-nano interfaces of AgNPs and Ag2SNPs with HSA: mechanistic insights into protein-nanoparticle complexation and their effects on antibacterial activity |
| **P-008** | Subhajit Dan | IIT Kanpur | Unraveling the Dynamics of Interfacial Water at Electrified Graphene Surfaces through Theoretical Two-Dimensional Vibrational Sum-Frequency Generation Spectroscopic Study |
| **P-009** | Hrishika Dey | IIT (ISM) Dhanbad | Targeting Plasmodium falciparum Calcium-Dependent Protein Kinases: Opportunities for Novel Antimalarial Therapeutics |
| **P-010** | Sampriti Ghosh | SINP Kolkata | Lanthanide-Doped Gold Nanocluster Assemblies Exhibiting Dopant-Directed Size Evolution, Symmetry-Controlled Eu3+ Emission, and Selective Chemical Recognition |
| **P-011** | Mansi Gupta | Shiv Nadar University, Delhi-NCR | Ultrafast Hole Transport via Host–Guest Binding in Perovskite Nanocrystals: Experiment and Theory |
| **P-012** | Injamamul Haque | Presidency University, Kolkata | Tuning Hydrogen Abstraction Reactivity of Fe(IV)=O Corrole Complexes through Local Electric Fields |

| No. | Presenter | Affiliation | Title |
|---|---|---|---|
| **P-013** | Krishna | SVNIT Surat | Interaction Mechanism in LiH-nNH3BH3 (n=1-2) Complexes: Insights from Vibrational and Chemical Shielding Analysis |
| **P-014** | Kajal Kumbhakar | | Enhanced Charge Transport Through Ion Networks in Highly Concentrated LiSCN-polyethylene Carbonate Solid Polymer Electrolytes |
| **P-015** | Rimi Kundu | IISER Kolkata | Unravelling Magnesium Dependent Conformational Switching in Flaviviral RNA |
| **P-016** | Sankar Maity | NIT Rourkela | Glucose-Mediated Stabilization of Ethanol-Stressed Lipid Bilayers: Insights from Molecular Dynamics |
| **P-017** | Srimila Mallick | IIT Guwahati | Enhanced Inhibition of Amyloid β Oligomerisation by a Coumarin- Chalcone Hybrid over a Coumarin Derivative |
| **P-018** | Sourav Manna | IIT Kharagpur | Thermal and Light-Induced Catalysis in Pyridoxal 5′-Phosphate-Dependent Aspartate Aminotransferase |
| **P-019** | Balaram Mondal | IIT Guwahati | Ligand-Specific Binding Mechanism and Dynamic Modulation of Auxin Receptor TIR1 by Structurally Diverse Auxin Analogs |
| **P-020** | Anwesha Mukherjee | University of Kalyani | Bile salt incorporated vesicles – Modulation of their characteristics and fluorescence probing of their microenvironment |
| **P-021** | Rik Niranjan Mukherjee | SNBNCBS Kolkata | Continuous Time Random Walk in Acetamide Deep Eutectics: Emergence of Dynamical Separation of Particles as Glass Transition is Approached |
| **P-022** | Sutanu Mukhopadhyay | SNBNCBS Kolkata | OrFeaUs: Harmonic-Percussive Source Separation (HPSS)-based Feature Extraction from Molecular Dynamics Trajectories |
| **P-023** | Tamisra Pal | | Benchmarking Generative-AI-driven sampling of protein conformational landscapes |
| **P-024** | Anuvab Panda | IIT Kanpur | Dynamical quantum chaos in one-dimensional transverse field Ising model |
| **P-025** | Subir Podder | Raiganj University | A Novel Naphthofluorescein Derivative as an Antioxidant for Biodiesel: Integrated DFT, Molecular Dynamics, and Machine Learning Insights |
| **P-026** | Priya | IIT Kanpur | Green’s function formalism for kinetic energy density functional for atoms. |
| **P-027** | Madhumita Rano | Ashoka University, Sonipat | State-specific computations of vibrational and electronic excited states |
| **P-028** | Rishu | IIT Delhi | Collective Reaction Dynamics during Ligand Exchange in Atomically Precise Metal Nanoclusters |
| **P-029** | Saswata Roy | Azim Premji University, Bengaluru | Challenges in scalable excited state semiclassical dynamics |
| **P-030** | Gourab Saha | IACS Kolkata | Quasiclassical Reaction Dynamics: Diels−Alder versus hetero-Diels−Alder Reactions |
| **P-031** | Dibyamanjaree Samant | IISER Kolkata | Mapping the sgRNA-induced Cas9 conformational activation pathways integrating structure-based modeling and VAMPnet |
| **P-032** | Arka Prava Sarkar | SNBNCBS Kolkata | Investigating the nature of anion migration in halide perovskites in presence of single layer graphene |
| **P-033** | Mokshi Sharma | Central University of Jammu | Variational Framework for the Calculation of Vibrational States and Thermal Properties with the Effective Harmonic Oscillator |
| **P-035** | Ananya Sinha | IACS Kolkata | Accelerating Selected Configuration Interaction with Restricted Boltzmann Machine |

| No. | Presenter | Affiliation | Title |
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| **P-036** | Vishal Yadav | IISER Kolkata | From Cause to Consequence: Explainable AI-Guided Reaction Coordinate Reveals Thermodynamic and Kinetic Mechanisms of Nucleic Acid Base-Flipping |

## POSTER SESSION B — Day 2, Thursday 29 October 2026, 16:15 – 17:45 (Session 11)

| No. | Presenter | Affiliation | Title |
|---|---|---|---|
| **IP-08** | Barun Kumar Maity | SINP Kolkata | Specific DNA Recognition Induces Folding Transition of the Oncogenic Fusion Protein PAX3FOXO1 |
| **IP-09** | Debashree Ghosh | IACS Kolkata | Title to be announced |
| **IP-10** | Padmaja Prasad Mishra | SINP Kolkata | Exploring Toehold-Independent DNA Strand Displacement through Parallel Triplex Formation |
| **IP-11** | Sudip Das | SINP Kolkata | Physics-Informed Machine Learning for Enzyme Catalysis |
| **IP-12** | Amit K. Paul | Bose Institute, Kolkata | Mode-Dependent Vibrational Energy Flow in Hydrogen-Bonded Dimers: A Unified Mechanism for Intramolecular Vibrational Energy Redistribution and Vibrational Predissociation |
| **IP-13** | Harun Al Rasid Gazi | Aliah University, Kolkata | Decoding the Molecular Interactions of Thymol, Thymoquinone, and Lawsone with DNA and Protein: Experimental & Computational Approach |
| **IP-14** | Rajarshi Chakrabarti | IIT Bombay | Active Polymers in Porous Media: Activity-Driven Conformational and Transport Dynamics |
| **IP-15** | Surajit Maity | IIT Hyderabad | Quantifying the Intrinsic Energetic Thresholds of Solvent-Driven Excited-State Proton Transfer via Isotopic Tunneling Quenching |
| **IP-16** | Tushar Kanti Mukherjee | IIT Indore | Regulation of Protease Activity of Papain via Biomolecular Condensation and Its Reactivation by L-Cysteine |
| **IP-17** | Arijit Kumar De | IISER Mohali | Unveiling Time-Dependent Vibronic Coupling by Third- and Fifth-order Raman Spectroscopy |
| **IP-27** | Abhijit Dan | MAKAUT, West Bengal | Interfacially Engineered Liquid Crystal Microdroplets: A Robust Platform for Label-Free Optical Biosensing |
| **P-037** | Md Asif Amin | Suri Vidyasagar College | Coronene Based Two Dimensional Metal-Organic Framework with High Spin-Filtering Efficiency |
| **P-038** | Nidhi Arya | IIT Jodhpur | Unravelling the $Mg^{2+}$-Dependent Structural Dynamics of DNA Holliday Junctions at the Single-Molecule Level |
| **P-039** | Indrani Bhattacharya | SNBNCBS Kolkata | Ascorbic Acid Modulates the Hydration Dynamics of Urea and Glucose: Insights from Terahertz Spectroscopy and Molecular Dynamics Simulations |
| **P-040** | Sayari Bhattacharya | SNBNCBS Kolkata | The Fibril Tip Sets the Route, Not Just the Destination: Polymorph-Specific Kinetics of Aβ 42 Monomer Incorporation |
| **P-041** | Sakshi Chawla | IIT Gandhinagar | Tailoring Herzberg-Teller Vibronic Interactions in Tetrapyrrole-Based Systems |
| **P-042** | Shrestha Chowdhury | SNBNCBS Kolkata | Structural Deviations from Ideality in Alcohol + Water Azeotropes: Molecular Scale Investigations |
| **P-043** | Kingshuk Das | IACS Kolkata | Salt-Mediated Molecular Recognition of the p53 C-terminal Domain |
| **P-044** | Krishnandu Dey | NIT Meghalaya | Photoswitch activity of an Acyl-hydrazone dimer: A non-adiabatic dynamics study. |
| **P-045** | Priya Dey | Kyoto University, Japan | Pepper-Mill Protein Dynamics: Decoding the Mechanism of MSPA for ssDNA Translocation |

| No. | Presenter | Affiliation | Title |
|---|---|---|---|
| **P-046** | Sk Samsul Ghaus | IISER Kolkata | Engineering HLCT-AIE Active Molecule for Efficient Flexible Optical Waveguides: Photophysics, Mechanics, and Optoelectronic Integration |
| **P-047** | Oushnik Sen Gupta | SINP Kolkata | Monitoring enzyme activity on the surface of a self-referenced near-white light emitting assembly of atomic clusters. |
| **P-048** | Subhabrata Hazra | IACS Kolkata | Intermediate Relaxation Timescale in Bulk Water: A Temperature-Dependent Investigation |
| **P-049** | Tathagata Islam | SINP Kolkata | Supramolecular assembly-driven emergent chirality with simultaneous induction and ligand inversion in atomic clusters |
| **P-050** | Vijay Kumar | IIT Kharagpur | Determinant Sampling and Subspace Diagonalization for Strongly Correlated Molecular Systems |
| **P-051** | Subrata Kundu | IIT Jodhpur | $Mg^{2+}$-Dependent Dynamics and Kinetics of a DNA Holliday Junction Studied by Single-Molecule FRET. |
| **P-052** | Himashree Mahanta | Assam Kaziranga University | Chemical Dynamics Simulations of Benzene-Hexachlorobenzene Complexes: Association, Ensuing Dissociation, and Comparison with Related π–π Molecular Complexes |
| **P-053** | Koushik Makhal | IIT Hyderabad | A potential gated formic acid oxidation reaction on the Rh-Cr dual-atom catalyst water interface |
| **P-054** | Apurba Manna | IIT Kharagpur | Unravelling the Microscopic Properties of a Globular Protein in Aqueous Deep Eutectic Solutions using Molecular Simulations |
| **P-055** | Archi Mishra | NIT Karnataka, Surathkal | Effect of membrane composition on the structural dynamics of Amyloid Precursor Protein(APP) |
| **P-056** | Piyali Mukherjee | IIT Bombay | A Theoretical Investigation of the Encapsulation of Small Molecules within Coacervates of Elastin-like Polypeptides |
| **P-057** | Venkatesh N | IIT Patna | Growth of Low-Density Liquid Domains Drives Fickian, Yet Non-Gaussian Diffusion in Supercooled Water |
| **P-058** | Rishabh Kumar Pandey | CUHK Shenzhen | Vibronic Coupling Effects in the Photoelectron Spectra of LiB6- |
| **P-059** | Roja Satvika Puvvada | GITAM University | Unravelling the role of aqueous choline chloride-polyol deep eutectic solvents in protein stability |
| **P-060** | Rajendra Rath | NIT Rourkela | Emergent Mixed Crowding Effect on Protein Dimerization |
| **P-061** | Indranil Roy | IISc Bengaluru | Anisotropic quantum beats report room temperature functional vibronic couplings in light-harvesting aggregates |
| **P-062** | Sudhriti Roy | IIT Jodhpur | Engineering Heterotypic Biomolecular Condensates with Synthetic Peptides for Controlled Spatial Organization and Liquid-like Nature |
| **P-063** | Gourab Saha | IIT Kharagpur | Molecular Insights into the Aβ-42 Tetramer Induced Disruptive Effects on a Model Phospholipid Membrane |
| **P-064** | Koushik Saha | IACS Kolkata | Quantum Dynamics and Mean Square Displacement of a Nonlinear Brownian Oscillator |
| **P-065** | Dipayan Sarder | IIT Kharagpur | Microscopic Understanding of DNA Groove Binding by Hydrophilic and Hydrophobic Ionic Liquids |
| **P-066** | Arghya Sen | University of Rostock, Germany | CsMnBr3 Nanocrystals: Excitation-Dependent Photophysics and Curious Case of Carrier Dynamics from Picoseconds to Microseconds |
| **P-067** | Pallav Sengupta | IIT Guwahati | Distal residue mutations dictate substrate binding in MtbThyX |

| No. | Presenter | Affiliation | Title |
|---|---|---|---|
| **P-068** | Deepak Sharma | IIT Jodhpur | Dual Effect of RNA-like Polyelectrolytes on Stability and Dynamics of Biomolecular Condensates: A Tale of Competitive Short- and Long-Range Interactions. |
| **P-069** | Vishnu Sudarsanan | IISER Pune | Li2 O Vacancy-Induced Modulation of Oxygen Redox in Li2MnO3 Surfaces: A First Principles Study |
| **P-105** | Titas Kumar Mukhopadhyay | NIT Jamshedpur | Self-Assembled PTCDA Coatings on Graphene Profoundly Improve the Biocompatibility Toward Genetic Materials |
| **P-106** | Debasish Barman | Kyushu University, Japan | Precise Control of Charge-Transfer for Triplet Harvesting: Functional Molecular Designs for High-Efficiency Narrowband TADF-OLEDs |

## POSTER SESSION C — Day 3, Friday 30 October 2026, 16:45 – 18:15 (Session 17)

| No. | Presenter | Affiliation | Title |
|---|---|---|---|
| **IP-18** | Bhabani S. Mallik | IIT Hyderabad | Multiscale Modeling of Electrochemical Catalysis and Energy Storage Systems via Molecular Dynamics Simulations |
| **IP-19** | Ranjit Biswas | SNBNCBS Kolkata | Conductivity in Battery Electrolyte Systems: Impact of Ultrafast Solution Dynamics, Ion-Ion Motional Correlations, and Solution Heterogeneity |
| **IP-20** | Suman Chakrabarty | SNBNCBS Kolkata | Navigating Pathways in Free Energy Landscape: Adaptive (Machine) Learning Strategies Towards Rare Event Kinetics |
| **IP-21** | Souvick Biswas | IISER Thiruvananthapuram | Resonantly Stabilized Free Radical Mediated Mass Growth: Indenyl Chemistry as the Common Thread of Planar and Curved PAHs |
| **IP-22** | Anup Ghosh | Bose Institute, Kolkata | Beyond the Sulfoxide Oxygen: NMR Spectroscopy and Machine-Learned Dynamics Reveal α-C–H···Ow Hydrogen Bonding in Dimethyl Sulfoxide-Water |
| **IP-24** | Padmabati Mondal | IISER Tirupati | A multiscale study in search of photoswitchable drug for serotonin receptors |
| **IP-25** | Anirban Panda | J. K. College, Purulia | Addition Reaction Networks and Partial Ordering of Phosphorus–Sulfur Binary Compounds: A Theoretical Study |
| **IP-26** | Subhadip Ghosh | NISER Bhubaneswar | Spatially Resolved Energy Transfer in Hexagonal Cs4PbBr6 Microdisks Revealed by Single-Particle Spectroscopy |
| **P-070** | Farhin Akhtar | RGIPT Sivasagar | Multifunctional Plasmonic Fluorescent Nanocomposites for Ultrasensitive Molecular Sensing, Antimicrobial Defense and High Resolution Bioimaging |
| **P-072** | Anupaul Baruah | Dibrugarh University | A Light-weight Monte Carlo Simulation Based In Silico Protein Design Method for Mutation Induced Fold Switch |
| **P-073** | Sreyan Bhowmick | SNBNCBS Kolkata | Ligand-driven entropic collapse directs mechanical strain to facilitate BirA dimerization |
| **P-074** | Ipsita Chakraborty | S. M. College, Bhagalpur | Unravelling PNIPAM Hysteresis: A Solvent Relaxation NMR Study of the Globule-to-Coil Transition |
| **P-075** | Samsung Raja Daimari | IIT Guwahati | Gas and Aqueous Phase Oxidation of a Series of Lactones: Initiated with OH Radicals |
| **P-076** | Supritam Datta | IIT Indore | A Sustainable and Catalyst-Free Direct Photochemical Route for the Conversion of Amines to Imines via C–N Coupling Reaction Under Ambient Conditions |
| **P-077** | Pradipta Dey | IACS Kolkata | Effect of Protein on Photoisomerization of Bacteriophytochrome |

| No. | Presenter | Affiliation | Title |
|---|---|---|---|
| **P-078** | Parineeta Gogoi | IISc Bengaluru | Tailoring Quantum Dot Properties for Quantum Science: Influence of Charge, Energy Transfer and Anisotropy on Exciton Dynamics in Quantum dots and their Assemblies |
| **P-079** | Yugant PM Guddha | IIT Guwahati | Unveiling the Influence of Biomolecules on the Relaxivity of Mn(II) Based Contrast Agents: A Molecular Dynamics Study |
| **P-080** | Aanchal Hurmade | IISc Bengaluru | Raman Spectroscopic Investigation of Biochemical Changes in Imiquimod-Induced Psoriasis Model. |
| **P-081** | Mira Jhawar | IIT Guwahati | From Core Insertion to Surface Binding: A Molecular Dynamics Study of hCT10-22 Fragment Behavior on Different Lipid Membranes |
| **P-082** | Neha Kundara | Shiv Nadar University, Delhi-NCR | Investigation of the interactions of influenza A virus-encoded miRNA with human PCBP2 mRNA and Agronaute-2 protein at the single-molecule level |
| **P-083** | Abhiroop Lahiri | IACS Kolkata | Committor-Based Detection of Post-Transition-State Bifurcations from Enhanced-Sampling Simulations |
| **P-084** | Dibyendu Maity | SNBNCBS Kolkata | Quantitative Pathway-Resolved Kinetics from Neural Network-Guided Weighted Ensemble Simulations |
| **P-085** | Abdul Aziz Mandal | SNBNCBS Kolkata | Mechanistic Role of Different Functional Groups in Inhibiting Tryptophan Self-Assembly: Comparison between Aromatic and Nonaromatic Hydroxyl Moiety-Rich Chemical Compounds |
| **P-086** | Anurag Maurya | IISc Bengaluru | Stochastic Thermodynamics of a Non-Markovian Random Process |
| **P-087** | Anandita Mitra | IIT Guwahati | Therapeutic Targeting of Cancer-Associated PKM2 Mutants by Shikonin via Mutation-Induced Pocket Recognition |
| **P-088** | Mainak Mukherjee | Sri Sathya Sai Institute of Higher Learning | Computational Investigation of Through-Bond and Through-Space Charge Transfer for Reduced or Inverted ΔEST in TADF Emitters |
| **P-089** | Shreya Mukherjee | IIT (ISM) Dhanbad | Estimation of Absolute Protein–DNA Binding Free Energy using Streamlined Geometric Formalism |
| **P-091** | Somnath Paitandi | IIT Guwahati | Mg-Co-B Chemical Space for Mg-Ion Cathodes: Evolutionary Search, Machine Learning, and Chemical Bonding Analysis |
| **P-092** | Shivam Pandey | IACS Kolkata | Turning Heat into Electricity: Driving High ZT through Soft Phonon Engineering |
| **P-093** | Juriti Rajbangshi | University of Wisconsin-Madison, USA | Molecular Insights and Predictive Models for Solvent Selection in Lignin Valorization |
| **P-094** | Samarth Razdan | IIT Bhilai | Molecular Engineering of Antiaromatic Orangarin-based Sensitizers for High-Performance Dye-Sensitized Solar Cell Applications |
| **P-095** | Roshan | IIT Delhi | Interfacial engineering of waste derived Char/VO2 composites for stable electrochemical energy storage |
| **P-096** | Arnab Roy | IACS Kolkata | Anti-Kasha-Type Spin-Vibronic Dynamics Drive Photoinduced Ni–H Umpolung in Ni-pincer complex |
| **P-097** | Debasish Saha | IISER Bhopal | Non-Markovian escape under stochastic resetting |
| **P-098** | Adyasa Sahoo | IIT Guwahati | Structural Responsiveness as the Key to Water’s Phase Behavior: Lessons from Machine-Learned mW and BOP Models. |
| **P-099** | Jagori Sarkar | IIT Kanpur | Decoding the Rate Determining Step in plasmidium SHMT to Guide Antimalarial Inhibition:A QM/MM Enhanced-Sampling Free-Energy Study |

| No. | Presenter | Affiliation | Title |
|---|---|---|---|
| **P-100** | Shaheerah Shahid | SNBNCBS Kolkata | Disentangling Heterogeneous Trajectory Ensembles by an Automated and Explainable Framework for Pathway Clustering and CV Identification |
| **P-101** | Mohd Ayaz Siddique | IIT Guwahati | Mechanistic Insights into Substrate Recognition by the Key Quorum-Sensing Regulator Protein SepM in Streptococcus mutans |
| **P-102** | Rahul Sutradhar | IACS Kolkata | Implicating the Role of Enhanced Oxyl Attribute in Facilitating Exceptional C−H Activation by a [Co2 III,IV(μ-O)2] Diamond-Core Complex: A Theoretical Perspective |
| **P-103** | Himanshu K. Tiwary | IIT Jodhpur | Synthesis and Characterization Study of Precursor-Controlled Multicolour Carbon Quantum Dots for Bioimaging and Drug Delivery Applications |
| **P-104** | Sumit Singha | University of North Bengal | Engineering Nanoscale Interfaces for Enhanced Förster Resonance Energy Transfer |